

PROJECT 1

DATA CLEANING

DecodeLabs Data Analytics Internship

Prepared By

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Tools Used

Microsoft Excel

Power Query

Microsoft Word

GitHub

Project Objective

To clean a raw dataset by handling missing values, removing duplicate records, and correcting date, numeric, and text formats.

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Project Overview

This project was completed as part of the DecodeLabs Data Analytics Internship Project 1. The objective of the project is to clean and prepare a raw dataset for analysis by handling missing values, removing duplicate records, and correcting inconsistent data formats.

The project was completed using Microsoft Excel Power Query, which provides an efficient and professional workflow for data transformation and cleaning.

Dataset Overview

The dataset was provided in Excel format and contained multiple columns with text, numeric, and date values. During the initial inspection, several data quality issues were identified, including missing values, duplicate records, inconsistent text formatting, and incorrect date formats.

Power Query profiling tools were used to inspect column quality, column distribution, and data consistency before starting the cleaning process.

Dataset Statistics

- Total Records: 1200
- Total Columns: 14
- Text Columns: 9 (including OrderID)
- Numeric Columns: 4
- Date Columns: 1
- Unique Order IDs: 1200

Dataset Import and Initial Inspection

The raw dataset was imported into Microsoft Excel and converted into a structured table format. Initial inspection of the dataset revealed multiple columns containing text, numeric, and date-based information.

During the initial review, formatting inconsistencies were identified in the Date column, which required correction before further processing in Power Query.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	OrderID	Date	CustomerID	Product	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemInCart	CouponCode	ReferralSource	TotalPrice
2	ORD200000	2023-01-04	C72649	Monitor	5	570.62	928 Main St	Debit Card	Shipped	TRK37947903	7	SAVE10	Instagram	2853.1
3	ORD200001	2024-08-23	C75739	Phone	2	151.35	823 Main St	Online	Shipped	TRK91186779	3	SAVE10	Referral	302.7
4	ORD200002	2024-02-27	C81728	Tablet	5	550.68	512 Main St	Credit Card	Cancelled	TRK42903982	8	FRESHIP	Email	2753.4
5	ORD200003	2023-10-15	C33540	Chair	1	273.19	275 Main St	Debit Card	Returned	TRK62788070	5	SAVE10	Facebook	273.19
6	ORD200004	2025-05-08	C81840	Printer	4	626.01	668 Main St	Online	Delivered	TRK29241424	8	SAVE10	Email	2504.04
7	ORD200005	2023-10-23	C37249	Phone	2	245.86	934 Main St	Credit Card	Shipped	TRK72976927	4	SAVE10	Instagram	491.72
8	ORD200006	2025-06-17	C83492	Laptop	1	664.42	986 Main St	Gift Card	Returned	TRK96417362	9	SAVE10	Facebook	664.42
9	ORD200007	2023-05-12	C41460	Monitor	5	149.55	706 Main St	Cash	Shipped	TRK78809193	9	FRESHIP	Facebook	747.75
10	ORD200008	2025-04-02	C26817	Phone	2	134.28	904 Main St	Gift Card	Cancelled	TRK61042692	2		Email	268.56
11	ORD200009	2023-11-21	C31946	Desk	4	509.38	102 Main St	Credit Card	Shipped	TRK33478363	6	SAVE10	Google	2037.52
12	ORD200010	2023-12-29	C43443	Tablet	5	625.97	333 Main St	Credit Card	Returned	TRK98859248	9	WINTER15	Instagram	3129.85
13	ORD200011	2024-02-17	C93861	Monitor	3	49.14	831 Main St	Online	Returned	TRK48234646	7	SAVE10	Email	147.42
14	ORD200012	2024-10-15	C38785	Monitor	2	180.5	179 Main St	Debit Card	Pending	TRK20419991	6	FRESHIP	Referral	361
15	ORD200013	2023-08-30	C88348	Laptop	3	201.49	903 Main St	Credit Card	Returned	TRK81913874	4	FRESHIP	Facebook	604.47
16	ORD200014	2023-03-27	C98474	Tablet	2	393.33	980 Main St	Debit Card	Pending	TRK79186539	3	SAVE10	Instagram	786.66
17	ORD200015	2023-07-17	C39416	Printer	1	473.96	942 Main St	Cash	Delivered	TRK54930938	3		Google	473.96
18	ORD200016	2023-01-24	C28976	Printer	2	533.81	914 Main St	Credit Card	Pending	TRK13194259	3	FRESHIP	Email	1067.62
19	ORD200017	2024-03-02	C60672	Tablet	1	423.4	382 Main St	Debit Card	Shipped	TRK11972116	3		Referral	423.4
20	ORD200018	2025-03-26	C84329	Desk	1	431.44	891 Main St	Gift Card	Pending	TRK65191045	3	SAVE10	Facebook	431.44
21	ORD200019	2023-01-23	C14552	Monitor	5	224	273 Main St	Cash	Returned	TRK59101365	9	SAVE10	Google	1120
22	ORD200020	2023-05-08	C54829	Chair	5	62.51	587 Main St	Cash	Pending	TRK16718651	9	WINTER15	Instagram	312.55
23	ORD200021	2024-11-17	C99023	Monitor	4	342.95	569 Main St	Cash	Delivered	TRK52646983	6		Google	1371.8
24	ORD200022	2023-01-29	C34416	Monitor	2	49.62	633 Main St	Online	Pending	TRK75937350	2	SAVE10	Google	99.24
25	ORD200023	2025-04-09	C42372	Phone	4	273.25	755 Main St	Debit Card	Cancelled	TRK36168083	5	FRESHIP	Referral	1093
26	ORD200024	2024-10-30	C17470	Laptop	2	230.95	956 Main St	Online	Delivered	TRK28775057	3	WINTER15	Referral	461.9
27	ORD200025	2024-07-24	C42577	Monitor	1	399.98	709 Main St	Debit Card	Delivered	TRK69423327	3	WINTER15	Email	399.98
28	ORD200026	2024-03-06	C21944	Laptop	5	59.55	296 Main St	Online	Delivered	TRK80276812	6		Referral	297.75
29	ORD200027	2024-02-22	C97593	Printer	1	472.74	881 Main St	Online	Shipped	TRK89117573	6	WINTER15	Google	472.74
30	ORD200028	2023-11-31	C96873	Phone	4	160.07	948 Main St	Online	Delivered	TRK65889530	6		Email	640.98

Figure 1: Raw Dataset in Excel Table Format

Importing Dataset into Power Query

After converting the dataset into a structured table, the data was imported into Microsoft Excel Power Query for further inspection and transformation.

Power Query provides advanced data cleaning and transformation capabilities, making it suitable for professional data preparation workflows.

The screenshot shows the Microsoft Excel interface with the Power Query Editor open. The main window displays a table with columns: OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemInCart, CouponCode, ReferralSource, and TotalPrice. The table contains 30 rows of data. The 'Query Settings' pane on the right shows the 'Table1' query with properties and applied steps. The 'Applied Steps' pane shows the 'Source' and 'Changed Type' steps. The 'Table Design' ribbon is visible at the top.

Figure 2: Dataset Imported into Power Query

Data Profiling and Quality Inspection

After importing the dataset into Power Query, data profiling tools were enabled to inspect data quality and identify inconsistencies within the dataset.

The Column Quality, Column Distribution, and Column Profile features were used to analyze valid values, empty values, duplicate entries, and overall column consistency before starting the cleaning process.

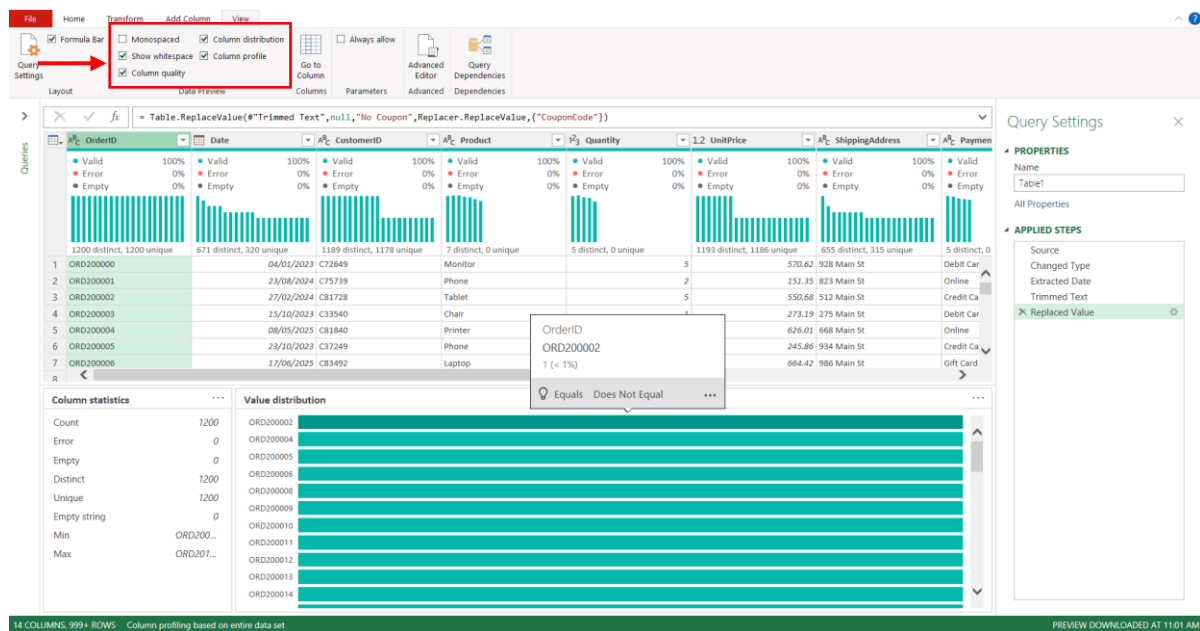


Figure 3: Data Profiling Results

Date Format Standardization

The Date column contained timestamp values with unnecessary time components. To improve consistency and prepare the dataset for analysis, the Date column was transformed into a clean date-only format using Power Query transformation tools.

This step ensured standardized date formatting across the dataset and eliminated unnecessary timestamp information.

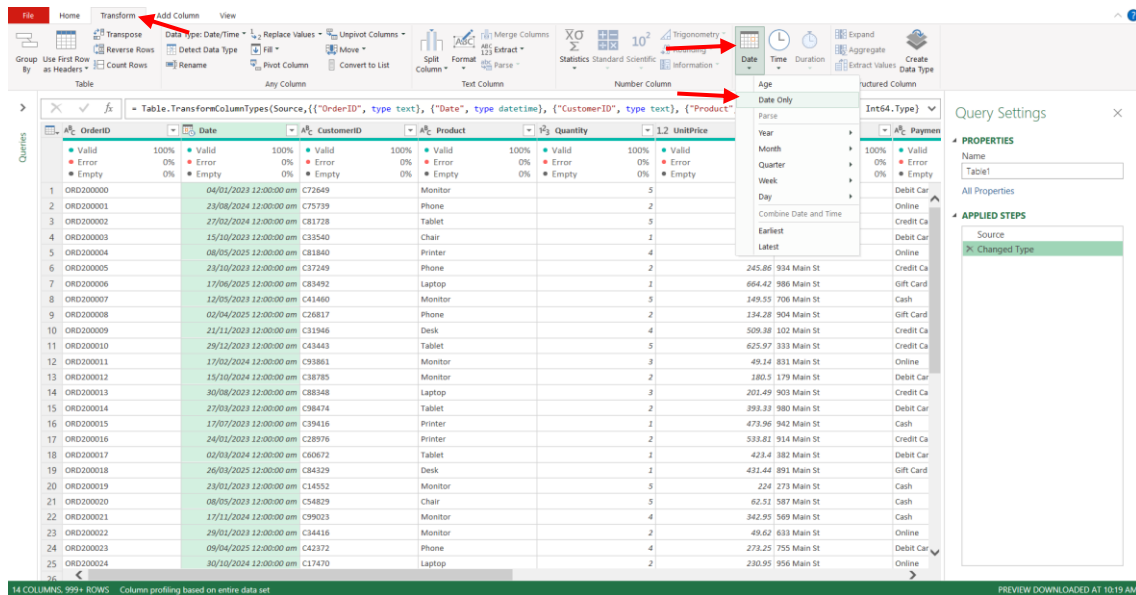


Figure 4: Date Format Standardization

Duplicate Record Validation

The OrderID column was analyzed to verify record uniqueness. Power Query statistics confirmed that all OrderID values were unique.

A total of 1200 distinct and 1200 unique Order IDs were identified, indicating that no duplicate records were present in the dataset.

This validation satisfied the project requirement of maintaining zero duplicate identifiers.

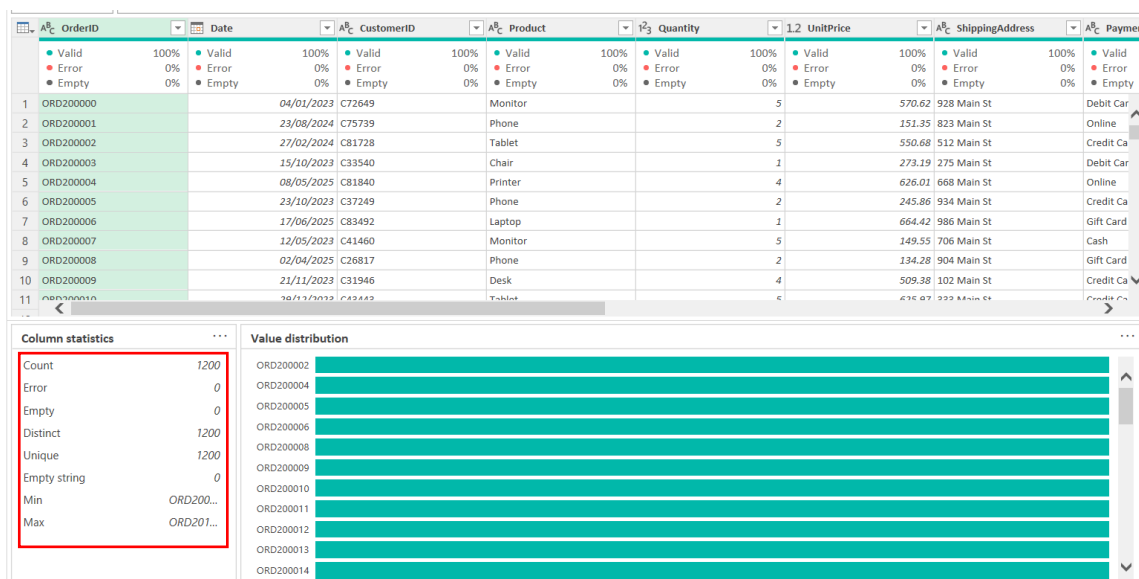


Figure 5: Duplicate Record Validation

Missing Values Analysis

During data profiling, missing values were identified in the CouponCode column. A total of 309 records contained null values, representing orders where no promotional coupon was applied.

Since the absence of a coupon code is a valid business scenario, missing values were replaced with the label "No Coupon" to improve data consistency and readability.

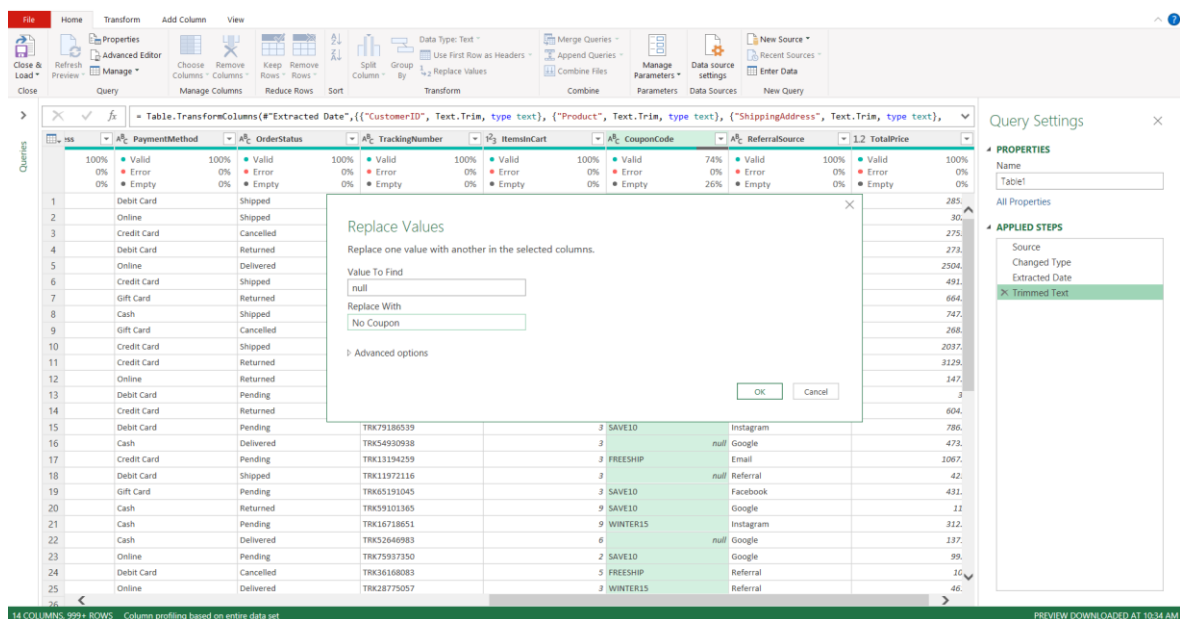


Figure 6: CouponCode Missing Values Handling

Data Type Validation

All dataset columns were reviewed to ensure appropriate data types were assigned. Text, numeric, and date columns were verified using Power Query's automatic type detection and validation features.

The Date column was assigned the Date data type, while Quantity, UnitPrice, ItemsInCart, and TotalPrice were maintained as numeric fields.

	Quantity	UnitPrice	ShippingAddress	PaymentMethod	OrderStatus	TrackingNumber	ItemsInCart	Cost
0%	Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty	100% Valid 0% Error 0% Empty
1		5	570.62 928 Main St	Debit Card	Shipped	TRK37947903		7 SAVI
2		2	151.35 823 Main St	Online	Shipped	TRK91186779		3 SAVI
3		5	550.68 512 Main St	Credit Card	Cancelled	TRK42903982		8 FREE
4		1	273.19 275 Main St	Debit Card	Returned	TRK62788070		5 SAVI
5		4	626.01 668 Main St	Online	Delivered	TRK29241424		8 SAVI
6		2	245.86 934 Main St	Credit Card	Shipped	TRK72976927		4 SAVI
7		1	664.42 986 Main St	Gift Card	Returned	TRK96417362		6 SAVI
8		5	149.55 706 Main St	Cash	Shipped	TRK78809193		9 FREE
9		2	134.28 904 Main St	Gift Card	Cancelled	TRK61042692		2
10		4	509.38 102 Main St	Credit Card	Shipped	TRK33478363		6 SAVI
11		5	625.97 333 Main St	Credit Card	Returned	TRK98859248		9 WIN
12		3	49.14 831 Main St	Online	Returned	TRK48234646		7 SAVI
13		2	180.5 179 Main St	Debit Card	Pending	TRK20419991		6 FREE
14		3	201.49 903 Main St	Credit Card	Returned	TRK81913874		4 FREE
15		2	393.33 980 Main St	Debit Card	Pending	TRK79186539		3 SAVI
16		1	473.96 942 Main St	Cash	Delivered	TRK54930938		3
17		2	533.81 914 Main St	Credit Card	Pending	TRK13194259		3 FREE
18		1	423.4 382 Main St	Debit Card	Shipped	TRK11972116		3
19		1	431.44 891 Main St	Gift Card	Pending	TRK65191045		3 SAVI
20								

Figure 7: Data Type Validation

Final Validation Results

After completing the data cleaning process, the dataset was validated against all project requirements.

Validation Table

Validation Check	Status
Missing Values	Passed
Duplicate IDs	Passed
Date Formatting	Passed
Data Types	Passed
Data Consistency	Passed

The final dataset was confirmed to be clean, consistent, and ready for further analytical processing.

Final Cleaned Dataset

After completing all data cleaning and validation activities, the transformed dataset was loaded back into Microsoft Excel. The final dataset contained standardized date values, validated data types, unique Order IDs, and properly handled missing values.

The cleaned dataset is now ready for data analysis, reporting, dashboard creation, and business decision-making processes.

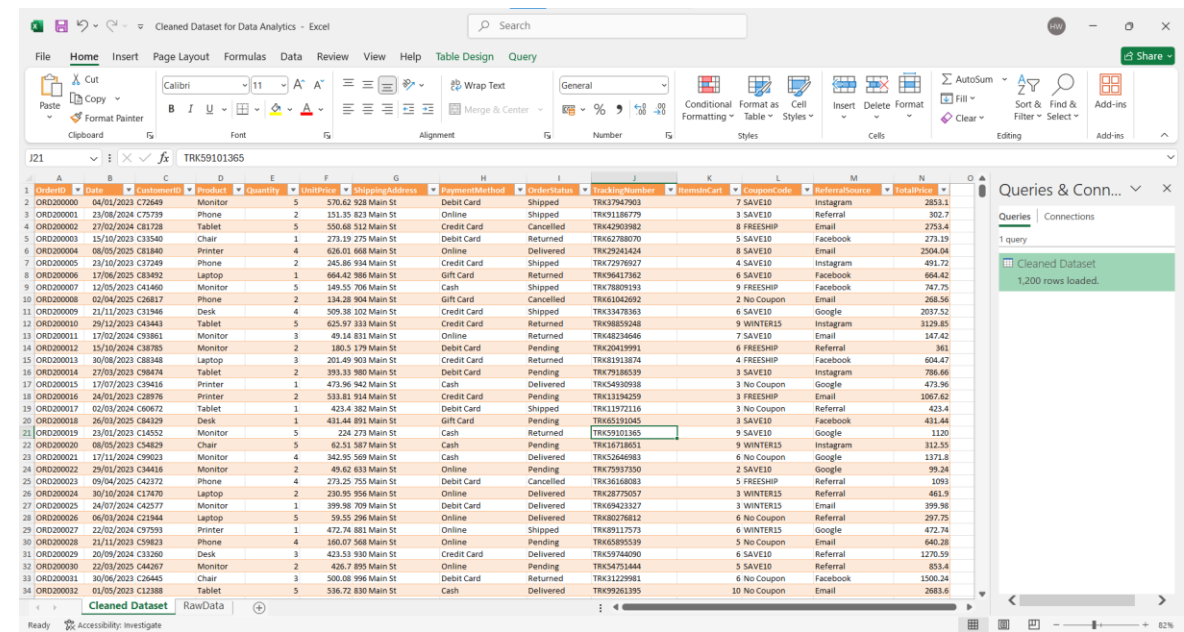


Figure 8: Final cleaned dataset loaded into Microsoft Excel after Power Query transformations

Transformation Summary Table

Step	Action Performed
1	Converted dataset into Excel Table
2	Imported data into Power Query
3	Enabled Data Profiling
4	Standardized Date Format
5	Validated Unique Order IDs
6	Replaced Null Coupon Codes with "No Coupon"
7	Verified Data Types
8	Final Validation Completed

Project Deliverables

- Raw Dataset (Dataset for Data Analytics.xlsx)
- Cleaned Dataset (Cleaned_Dataset.xlsx)

- Data Cleaning Report (.docx)
- Data Cleaning Report (.pdf)
- Power Query Transformation Steps
- Project Screenshots

Conclusion

This project provided practical experience in data cleaning and preparation using Microsoft Excel Power Query. The dataset was successfully inspected, validated, and standardized according to professional data quality practices.

Through this project, key data analytics skills such as data profiling, validation, transformation, and documentation were applied. The final dataset is now suitable for reporting, dashboard creation, and advanced analytical tasks.